

SOC-5811 Week 14: Transformations

Nick Graetz

University of Minnesota, Department of Sociology

12/1/2025

1/41



UNIVERSITY OF MINNESOTA

Driven to DiscoverSM



TRANSFORMATIONS

- ▶ How do we relax our assumptions around additivity and linearity?



TRANSFORMATIONS

- ▶ How do we relax our assumptions around additivity and linearity?
 - ▶ Additivity: add interactions.



TRANSFORMATIONS

- ▶ How do we relax our assumptions around additivity and linearity?
 - ▶ Additivity: add interactions.
 - ▶ Linearity: transform variables.



TRANSFORMATIONS

- ▶ How do we relax our assumptions around additivity and linearity?
 - ▶ Additivity: add interactions.
 - ▶ Linearity: transform variables.
- ▶ What does “relax” mean in this context?



TRANSFORMATIONS

- ▶ How do we relax our assumptions around additivity and linearity?
 - ▶ Additivity: add interactions.
 - ▶ Linearity: transform variables.
- ▶ What does “relax” mean in this context?
 - ▶ Rule of thumb: Inferences made with simpler models rely on *more* assumptions.



TRANSFORMATIONS

$$\text{Happiness}_i = \beta_0 + \beta_1 \text{Income}_i$$

Say I want to use this model to infer whether people with 150k are happier, on average, than people with 50k.

What assumption does this inference rely on?

TRANSFORMATIONS

What are "transformations?"

- ▶ We aren't talking about *shifting* or *rescaling* predictor variables, which changes our interpretation of the coefficients on those variables.



TRANSFORMATIONS

What are "transformations?"

- ▶ We aren't talking about *shifting* or *rescaling* predictor variables, which changes our interpretation of the coefficients on those variables.
- ▶ We are talking about changing the *functional relationship* between a predictor variable and the outcome.



TRANSFORMATIONS

- ▶ Log-transforming



TRANSFORMATIONS

- ▶ Log-transforming
- ▶ Polynomials

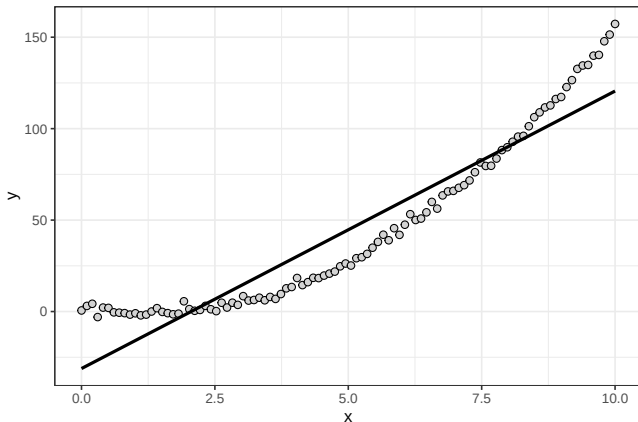


TRANSFORMATIONS

- ▶ Log-transforming
- ▶ Polynomials
- ▶ Splines

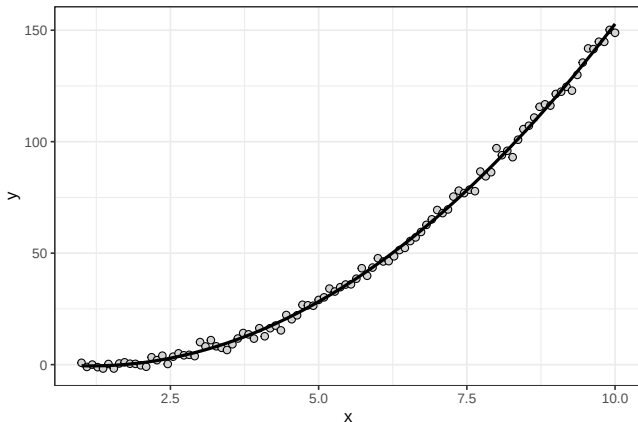
TRANSFORMATIONS

$$y_i = \beta_0 + \beta_1 X_i$$



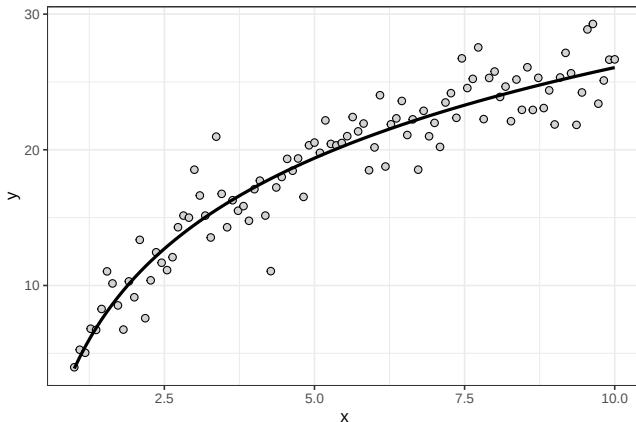
TRANSFORMATIONS

$$y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2$$



TRANSFORMATIONS

$$y_i = \beta_0 + \beta_1 \log(X_i)$$



TRANSFORMATIONS

$$y_i = \beta_0 + \beta_1 \log(X_i)$$

`summary(mod)`

```
##
## Call:
## lm(formula = y ~ log_x, data = d)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.8128 -1.1737  0.2431  0.9711  5.4042
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.8778      0.5343   7.258 9.36e-11 ***
## log_x         9.6341      0.3208  30.027 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.917 on 98 degrees of freedom
## Multiple R-squared:  0.902, Adjusted R-squared:  0.901
## F-statistic: 901.6 on 1 and 98 DF, p-value: < 2.2e-16
```

$$\beta_1 = 9.6$$

TRANSFORMATIONS

$$y_i = \beta_0 + \beta_1 \log(X_i)$$

$$\beta_1 = 9.6$$

Interpretation:

- ▶ A one-unit increase in $\ln(X)$ is associated with a 9.6 increase in Y .

TRANSFORMATIONS

$$y_i = \beta_0 + \beta_1 \log(X_i)$$

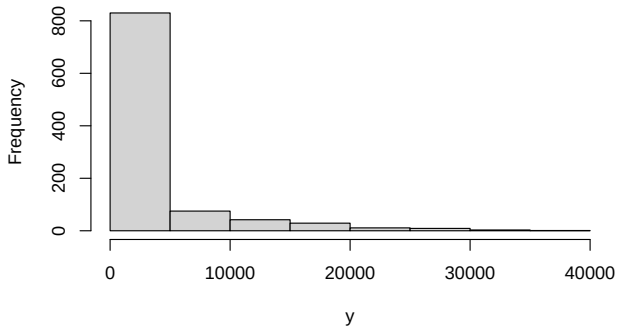
$$\beta_1 = 9.6$$

Interpretation:

- ▶ A one-unit increase in $\ln(X)$ is associated with a 9.6 increase in Y .
- ▶ A 1% increase in X is associated with a .096 increase in Y .

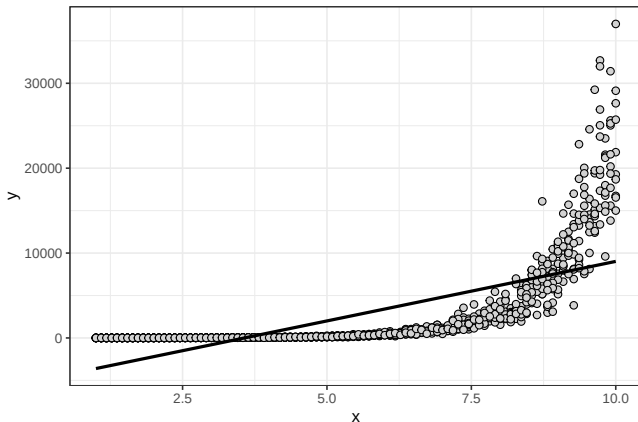
TRANSFORMATIONS

Histogram of y



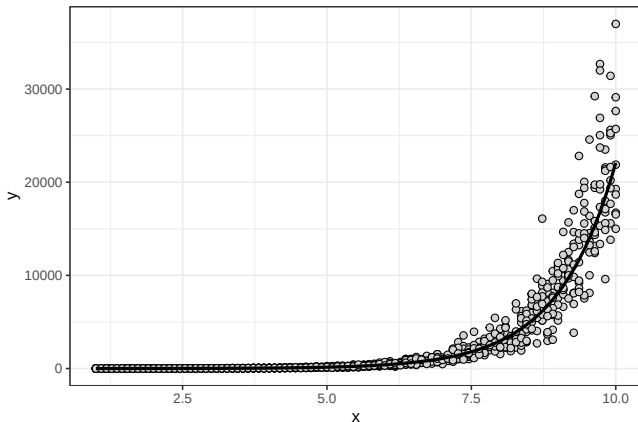
TRANSFORMATIONS

$$y_i = \beta_0 + \beta_1(X_i)$$



TRANSFORMATIONS

$$\log(y_i) = \beta_0 + \beta_1(X_i)$$



Why would we do this?

- ▶ Avoid negative predictions for an continuous outcome variable that is bounded at 0.

TRANSFORMATIONS

Why would we do this?

- ▶ Avoid negative predictions for an continuous outcome variable that is bounded at 0.
- ▶ If we think the relationship between X and Y is multiplicative rather than additive.



TRANSFORMATIONS

- ▶ **Only the dependent/response variable is log-transformed.** Exponentiate the coefficient. This gives the multiplicative factor for every one-unit increase in the independent variable.



TRANSFORMATIONS

- ▶ **Only the dependent/response variable is log-transformed.** Exponentiate the coefficient. This gives the multiplicative factor for every one-unit increase in the independent variable.
- ▶ **Only independent/predictor variable(s) is log-transformed.** Divide the coefficient by 100. This tells us that a 1% increase in the independent variable increases (or decreases) the dependent variable by (coefficient/100) units.



TRANSFORMATIONS

- ▶ **Both dependent/response variable and independent/predictor variable(s) are log-transformed.**
Interpret the coefficient as the percent increase in the dependent variable for every 1% increase in the independent variable.



TRANSFORMATIONS

- Pick up here on Wednesday



TRANSFORMATIONS

- Log-transforming



TRANSFORMATIONS

- ▶ Log-transforming
- ▶ Polynomials

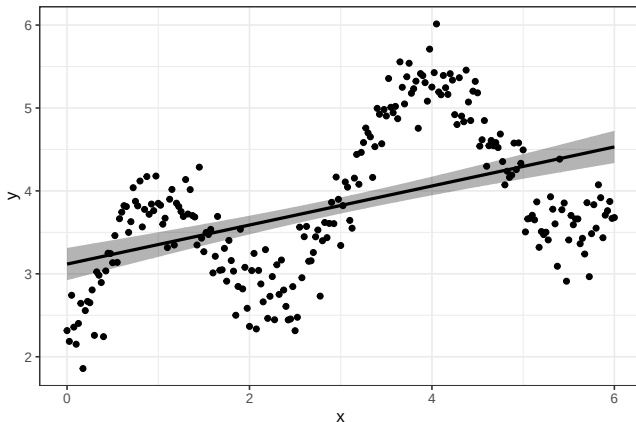


TRANSFORMATIONS

- ▶ Log-transforming
- ▶ Polynomials
- ▶ **Splines**

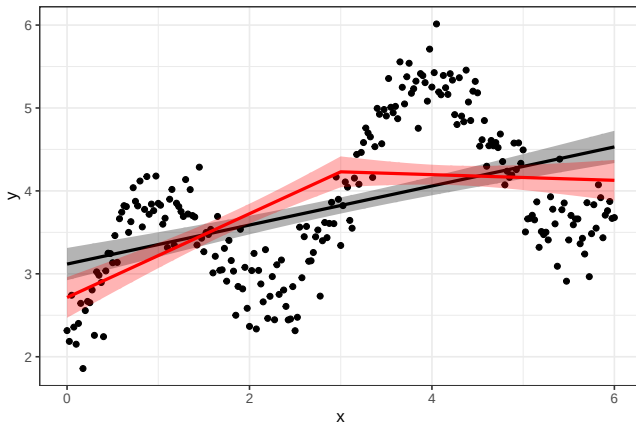
TRANSFORMATIONS

```
mod <- lm(y~x, data=d)
```



TRANSFORMATIONS

```
mod <- lm(y~lspline(x,knots=3), data=d)
```

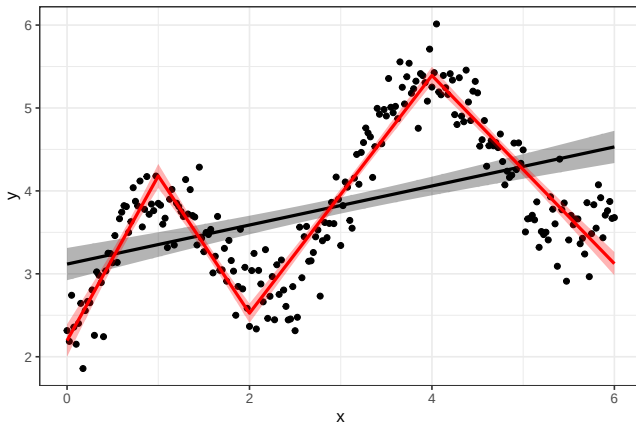


TRANSFORMATIONS

```
##
## Call:
## lm(formula = y ~ lspline(x, knots = 3), data = d)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.6634 -0.5753 -0.0236  0.6164  1.8192
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      2.71471    0.12506   21.707 < 2e-16 ***
## lspline(x, knots = 3)1  0.50515    0.06110    8.268 9.63e-15 ***
## lspline(x, knots = 3)2 -0.03417    0.06110   -0.559  0.576
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7377 on 238 degrees of freedom
## Multiple R-squared:  0.293, Adjusted R-squared:  0.2871
## F-statistic: 49.32 on 2 and 238 DF, p-value: < 2.2e-16
```

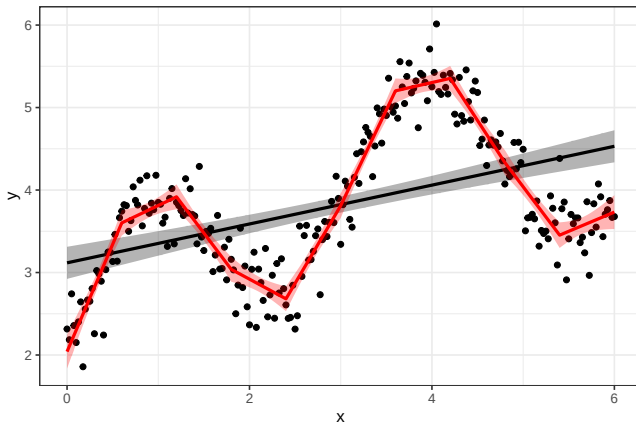
TRANSFORMATIONS

```
mod <- lm(y~lspline(x,knots=c(1,2,4)), data=d)
```



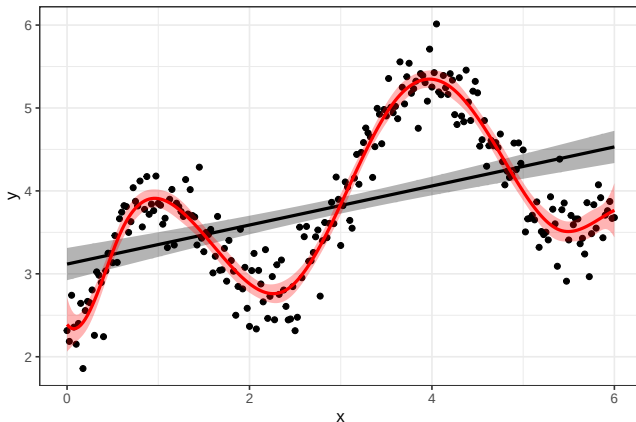
TRANSFORMATIONS

```
mod <- lm(y~pspline(x,degree=1), data=d)
```



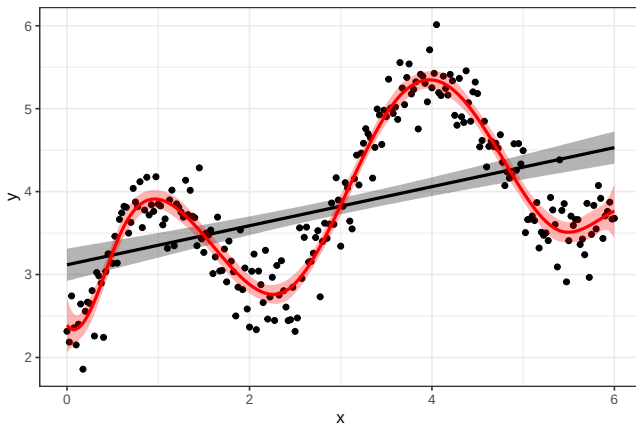
TRANSFORMATIONS

```
mod <- lm(y~pspline(x,degree=3), data=d)
```



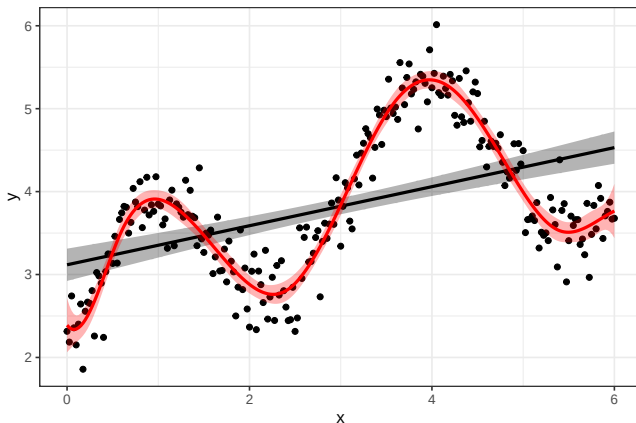
TRANSFORMATIONS

Penalized cubic splines



TRANSFORMATIONS

Which model do we like? Linear or penalized cubic spline?



TRANSFORMATIONS

Cons of spline: interpretation of coefficients

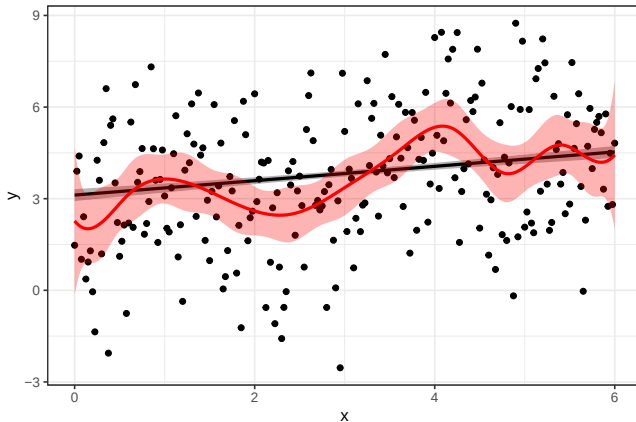
```
##
## Call:
## lm(formula = y ~ pspline(x, degree = 3), data = d)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.72148 -0.16581  0.00908  0.17742  0.85489
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      5.7273     1.9864   2.883  0.00431 **
## pspline(x, degree = 3)1  -4.5816     2.3090  -1.984  0.04843 *
## pspline(x, degree = 3)2  -1.7045     1.8754  -0.909  0.36438
## pspline(x, degree = 3)3  -1.7892     2.0465  -0.874  0.38291
## pspline(x, degree = 3)4  -2.7016     1.9614  -1.377  0.16975
## pspline(x, degree = 3)5  -3.2271     2.0060  -1.609  0.10907
## pspline(x, degree = 3)6  -1.9271     1.9822  -0.972  0.33197
## pspline(x, degree = 3)7  -0.4043     1.9951  -0.203  0.83958
## pspline(x, degree = 3)8  -0.2632     1.9883  -0.132  0.89481
## pspline(x, degree = 3)9  -1.3715     1.9930  -0.688  0.49204
## pspline(x, degree = 3)10 -2.4898     1.9935  -1.249  0.21295
## pspline(x, degree = 3)11 -1.8631     2.0195  -0.923  0.35722
## pspline(x, degree = 3)12 -1.8242     2.8043  -0.651  0.51602
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2807 on 228 degrees of freedom
## Multiple R-squared:  0.902, Adjusted R-squared:  0.8968
## F-statistic: 174.8 on 12 and 228 DF, p-value: < 2.2e-16
```

27/41



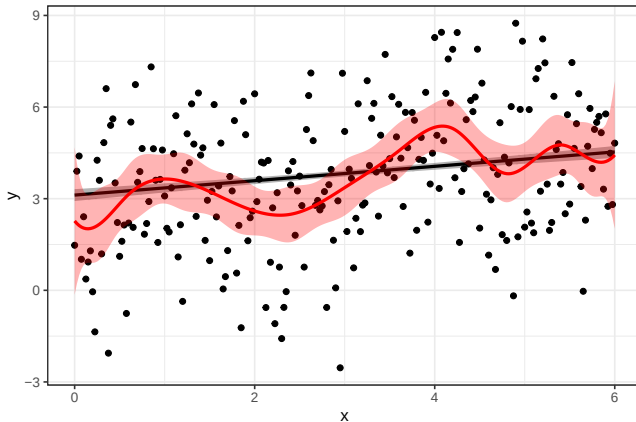
TRANSFORMATIONS

► Splines vs. linear



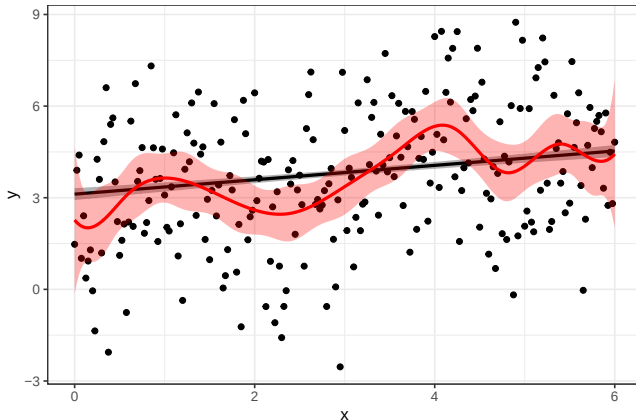
TRANSFORMATIONS

- What do we think? Which model do we prefer?



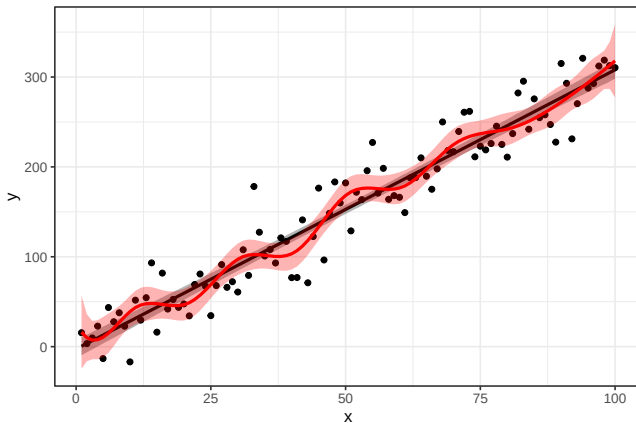
TRANSFORMATIONS

- What do we notice about the confidence intervals on the predictions?



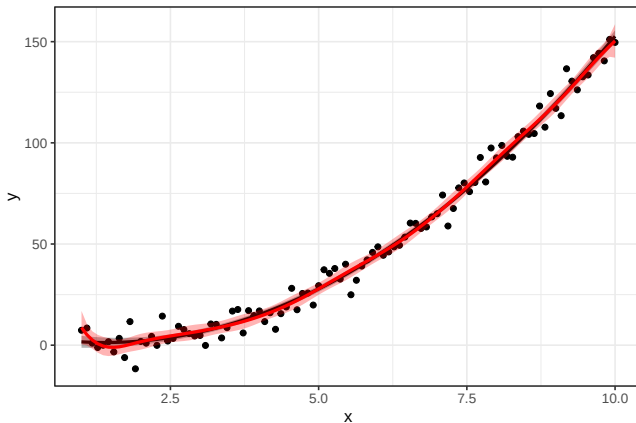
TRANSFORMATIONS

► Splines vs. linear



TRANSFORMATIONS

► Splines vs. polynomials



TRANSFORMATIONS

Causal inference: Transforming the treatment variable

- **Quiz:** I'm interested in the effect of Income (i.e., β_1), and Age is a potential confounder. Inference on β_1 requires more assumptions in which of these models?

$$\text{Happy}_i = \beta_0 + \beta_1 \text{Income}_i$$

$$\text{Happy}_i = \beta_0 + \beta_1 \text{Income}_i + \beta_2 \text{Age}_i$$

$$\text{Happy}_i = \beta_0 + \beta_1 S(\text{Income}_i) + \beta_2 \text{Age}_i$$

TRANSFORMATIONS

Causal inference: Transforming the treatment variable

- **Quiz:** I'm interested in the effect of Income (i.e., β_1), and Age is a potential confounder. Inference on β_1 requires more assumptions in which of these models?

Heterogenous treatment effects / Moderation

$$\text{Happy}_i = \beta_0 + \beta_1 S(\text{Income}_i) + \beta_2 \text{Age}_i$$

$$\text{Happy}_i = \beta_0 + \beta_1 \text{Income}_i + \beta_2 \text{Age}_i + \beta_3 (\text{Income}_i \times \text{Age}_i)$$

Causal inference: Transforming the confounder variable

- **Quiz:** I'm interested in the effect of Income (i.e., β_1), and Age is a potential confounder. Inference on β_1 requires more assumptions in which of these models?

$$\text{Happy}_i = \beta_0 + \beta_1 \text{Income}_i$$

$$\text{Happy}_i = \beta_0 + \beta_1 \text{Income}_i + \beta_2 \text{Age}_i$$

$$\text{Happy}_i = \beta_0 + \beta_1 \text{Income}_i + \beta_2 S(\text{Age}_i)$$

TRANSFORMATIONS

- ▶ **Rule of thumb:** Inferences made with simpler models rely on *more* assumptions.

TRANSFORMATIONS

- ▶ **Rule of thumb:** Inferences made with simpler models rely on *more* assumptions.
- ▶ **Quiz:** If we want to relax our assumptions of additivity and linearity as much as possible, why not just use splines for all continuous variables and include all interactions?



TRANSFORMATIONS

- ▶ **Rule of thumb:** Inferences made with simpler models rely on *more* assumptions.
- ▶ **Quiz:** If we want to relax our assumptions of additivity and linearity as much as possible, why not just use splines for all continuous variables and include all interactions?
- ▶ There are models out there besides the conventional linear model that basically do this!



TRANSFORMATIONS

- ▶ **Rule of thumb:** Inferences made with simpler models rely on *more* assumptions.
- ▶ **Quiz:** If we want to relax our assumptions of additivity and linearity as much as possible, why not just use splines for all continuous variables and include all interactions?
- ▶ There are models out there besides the conventional linear model that basically do this!
 - ▶ How do they handle the complexity vs. overfitting tradeoff? Cross-validation.



TRANSFORMATIONS

Why do we sometimes prefer simpler models?

- ▶ We want parameter estimates that are easy to interpret.



Why do we sometimes prefer simpler models?

- ▶ We want parameter estimates that are easy to interpret.
 - ▶ This is the weakest reason, because we can always just use conditional/marginal predictions for our inferences even if the model is very complicated in the parameters.

Why do we sometimes prefer simpler models?

- ▶ We want parameter estimates that are easy to interpret.
 - ▶ This is the weakest reason, because we can always just use conditional/marginal predictions for our inferences even if the model is very complicated in the parameters.
 - ▶ But it is also the reason that has historically driven the use of simpler models; it is very easy to estimate the linear regression coefficient.

Why do we sometimes prefer simpler models?

- ▶ We want parameter estimates that are easy to interpret.
 - ▶ This is the weakest reason, because we can always just use conditional/marginal predictions for our inferences even if the model is very complicated in the parameters.
 - ▶ But it is also the reason that has historically driven the use of simpler models; it is very easy to estimate the linear regression coefficient.
- ▶ We care about summarizing with average effects.

TRANSFORMATIONS

Why do we sometimes prefer simpler models?

- ▶ We want parameter estimates that are easy to interpret.
 - ▶ This is the weakest reason, because we can always just use conditional/marginal predictions for our inferences even if the model is very complicated in the parameters.
 - ▶ But it is also the reason that has historically driven the use of simpler models; it is very easy to estimate the linear regression coefficient.
- ▶ We care about summarizing with average effects.
- ▶ We can demonstrate that our assumptions (e.g., linearity) do not substantively impact our conclusions.

We have different techniques for **transformations** of continuous variables.

- Logs

REVIEW

We have different techniques for **transformations** of continuous variables.

- ▶ Logs
- ▶ Polynomials



REVIEW

We have different techniques for **transformations** of continuous variables.

- ▶ Logs
- ▶ Polynomials
- ▶ Splines



REVIEW

When do we think about **transforming** continuous variables or including **interactions**?

- ▶ **Predictive model**

When do we think about **transforming** continuous variables or including **interactions**?

- ▶ **Predictive model**

- ▶ Try everything and use cross-validation to see if it improves out-of-sample fit.

REVIEW

When do we think about **transforming** continuous variables or including **interactions**?

- ▶ **Predictive model**

- ▶ Try everything and use cross-validation to see if it improves out-of-sample fit.

- ▶ **Causal model**



REVIEW

When do we think about **transforming** continuous variables or including **interactions**?

- ▶ **Predictive model**

- ▶ Try everything and use cross-validation to see if it improves out-of-sample fit.

- ▶ **Causal model**

- ▶ Examining heterogeneous treatment effects (transform/interact the treatment variable).



REVIEW

When do we think about **transforming** continuous variables or including **interactions**?

- ▶ **Predictive model**

- ▶ Try everything and use cross-validation to see if it improves out-of-sample fit.

- ▶ **Causal model**

- ▶ Examining heterogeneous treatment effects (transform/interact the treatment variable).
- ▶ Relaxing assumptions about confounding pathways (transform/interact the confounder variables).

REVIEW

- ▶ Writing up a quantitative paper.



HOUSEKEEPING

- ▶ I need to move class next Monday (12/8) onto Zoom; I will send an invite link.



HOUSEKEEPING

- ▶ I need to move class next Monday (12/8) onto Zoom; I will send an invite link.
- ▶ The Student Rating of Teaching data collection window is open and closes next Wednesday (12/10); the link is on the menu on the left of Canvas.

